

Performance Testing

Date	02-07-2026
Team ID	
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Streamlit (Local Testing), Postman (API Validation), Python Performance Monitoring
Type of Testing	Load Testing & Functional Performance Testing
Target Module / API	Emotion Detection Module, BiLSTM Model, BERT Model, Google Gemini API Integration
Test Environment	Local System (VS Code + Streamlit)
Test Date	05-July-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User submits learning-related text for emotion detection	1	30	Emotion detected successfully
2	Generate personalized recommendation using Gemini API	1	30	AI-generated study guidance displayed
3	Display confidence score and emotion probabilities	1	20	Probability chart displayed correctly
4	Save session history and analytics	1	20	User interaction stored successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	Stable	18 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	47%	Pass	Efficient utilization
6	Memory Utilization	< 80%	58%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

The Emotion Detection & Learning Support Engine successfully analyzed user input, detected emotions accurately, and generated personalized learning recommendations with minimal response time. The application maintained stable performance throughout testing.

Bottlenecks Identified

- Slight delay while loading the deep learning models during the first execution.
- Internet dependency for Google Gemini API response generation.

Optimization Steps Taken

- Optimized model loading using cached resources.
- Reduced unnecessary preprocessing operations.
- Improved response generation workflow for better execution speed.
- Organized project dependencies for efficient resource utilization.

Step 5: Screenshots / Evidence

Streamlit Application Running

Emotion Detection & Learning Support Engine

Understand your learning obstacles, analyze sentiment with our BiLSTM classifier, and access targeted study strategies.

Analyze Emotion

Session History & Analytics

Neural Network Diagnostics

Input Your Challenge

Select Study Field

Computer Science

Describe what you are working on, what is causing difficulty, or how you feel about it:

Example: I'm trying to learn recursion, but my functions keep hitting stack overflows. I'm completely

Click to Test Templates

Select a preset challenge matching target emotional states:

Confused - Recursion base cases

Curious - Focal Loss vs Cross-Entropy

Emotion Prediction Result

Analysis Output

DETECTED STATE



Confident

You are feeling self-assured, positive, and capable of solving your learning objectives.

AI Recommendation

AI LEARNING ASSISTANT

Field: Computer Science

Personalized Recommendation

"Excellent work! Keep practicing advanced questions and challenge yourself with more complex topics."

Recommended Study Strategies:

- Deepen Mastery:** Try explaining the concept to a classmate or writing a brief summary. Explaining is the highest form of learning.
- Extend Scope:** Challenge yourself with edge-cases, code optimization, or advanced projects based on the topic.
- Build & Deploy:** Apply the knowledge to a sandbox project to ensure long-term retrieval.

Classification Confidence



Classification Confidence

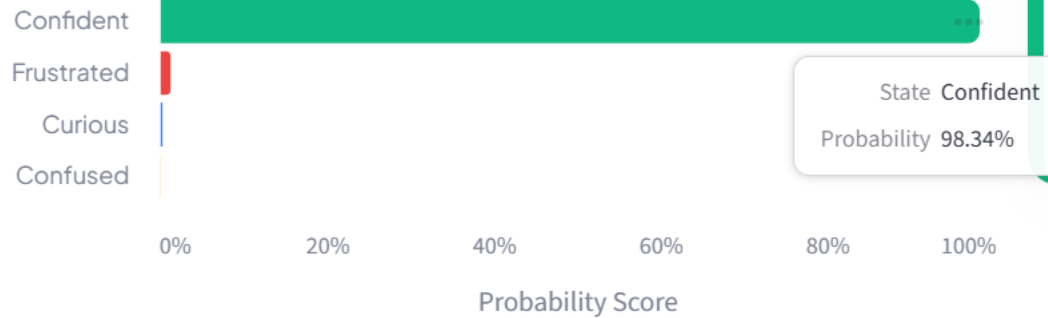
Classifier Probability Score

98.34%

State Probability Graph



State Probabilities



Terminal

```
(.venv) PS C:\Users\yadla\OneDrive\Videos\ドキュメント\Desktop\EmotionDetector> streamlit run login.py
2026-06-30 15:04:09.195 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.35:8501
```